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Identifying Cardiovascular Disease Risk with Machine Learning

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Identifying Cardiovascular Disease Risk with Machine Learning" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of our Industry Mentor. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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Finally, we thank our fellow researchers in the cardiovascular disease prediction and machine learning communities whose prior work laid the foundation upon which this research builds.

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ABSTRACT

The term "cardiovascular disease" (CVD) refers to a group of diseases that affect the circulatory system, which includes the heart and blood vessels that carry blood throughout the body. CVD is the reason for 30% of all the world's deaths, making it the leading cause of global mortality. Predicting CVD in a person is a tough and complex task that requires the analysis of multiple clinical and physiological parameters.

To improve the prediction accuracy, this study utilised the publicly available UCI Cleveland Clinic Heart Disease dataset comprising 303 patient records. Four machine learning classifiers were applied: K-Nearest Neighbour (KNN), Naive Bayes (NB), Logistic Regression (LR), and Decision Tree (DT). Two feature selection techniques — Singular Value Decomposition (SVD) and Linear Discriminant Analysis (LDA) — were implemented to enhance model performance by reducing noise and dimensionality.

The experimental results demonstrate that Naive Bayes with SVD feature selection achieved the highest accuracy of 88.50% in detecting whether a person is suffering from CVD or not. This accuracy surpasses all eight comparable prior studies on the same dataset. The results confirm that appropriate feature selection significantly improves classifier performance and offers a reliable tool for early-stage CVD prediction in clinical settings.

Keywords: CVD Detection, KNN, Logistic Regression, Decision Tree, Naive Bayes, Feature Selection, SVD, LDA.

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1. INTRODUCTION

Cardiovascular disease (CVD) is introduced as the principal global cause of mortality. This category encompasses various disorders affecting the cardiac muscle and the vascular system supplying crucial organs such as the heart and brain. In the early 20th century, CVD contributed to less than 10 percent of global deaths, but the scenario underwent a drastic shift, with the figure reaching 30 percent by 2001.

Notably, approximately 80 percent of CVD-related deaths transpire in low- and middle-income countries. CVD would emerge as the leading cause of global mortality and disability by 2020, primarily driven by its escalation in low- and middle-income nations. By 2001, CVD had assumed the leading cause of death in the developing world, mirroring its long-standing status in the developed world since the mid-1900s.

Detecting CVD is considered a tough task, but by detecting it in the early stage we can prevent death by a significant number. In England, around 62 million individuals are affected by CVD, while another 50 million are dealing with hypertension. CVD carries on as the No. 1 global cause of death, accounting for 17.3 million deaths per year — a number that is expected to reach 23.6 million by 2030. In recent years, the occurrence of traditional risk factors for atherosclerotic cardiovascular diseases has surged in populous developing nations like China and India. This rise has led to an increase in coronary and cerebrovascular events.

Early diagnosis plays a crucial role in providing timely and effective care for patients, impacting both individual health and the financial aspects of healthcare systems. However, current prediction strategies, especially in CVD, are often inefficient and time-consuming. Processing large amounts of information about a patient's heart is tedious even for people working in the health sector. Using machine learning to create a model which provides accurate and early detection will be very helpful in this scenario.

Machine learning is already successful in the diagnosis of breast cancer, skin cancer, and other diseases — and this research demonstrates it can be equally effective in cardiovascular disease prediction. The study proposes a systematic machine learning pipeline using four classifiers (KNN,

Naive Bayes, Logistic Regression, and Decision Tree) with two feature selection techniques (SVD and LDA) applied to the UCI Cleveland Clinic Heart Disease dataset.

1.1 Background

Cardiovascular diseases represent a spectrum of conditions affecting the heart and circulatory system. These include coronary artery disease, heart failure, arrhythmia, and hypertension. The global burden of these diseases has been escalating since the mid-20th century due to changing lifestyle patterns, increased sedentary behaviour, dietary changes, and rising obesity rates worldwide.

The challenge of CVD prediction lies in the complex interplay of multiple risk factors — age, sex, blood pressure, cholesterol levels, and electrocardiographic results — that together determine an individual's cardiovascular risk profile. Traditional risk stratification models, while useful, often fall short in capturing the non-linear relationships between these variables and disease outcomes.

1.2 Problem Statement

Despite significant advances in medical diagnostics, CVD remains the world's leading killer primarily because it is often detected too late. The current diagnostic process requires a cardiologist to manually interpret multiple clinical parameters, which is time-consuming, prone to human error, and inaccessible in resource-limited settings. There is a pressing need for an automated, accurate, and computationally efficient system that can predict CVD risk from standard clinical data.

1.3 Objectives

The primary objectives of this research are:

- To evaluate the performance of four machine learning classifiers — KNN, Naive Bayes, Logistic Regression, and Decision Tree — on the UCI Cleveland Heart Disease dataset.
- To investigate the impact of two feature selection techniques — SVD and LDA — on classifier accuracy.
- To identify the optimal classifier and feature selection combination for CVD prediction.
- To compare our results with existing literature and demonstrate superior performance.
- To provide a reliable machine learning tool for early-stage CVD detection that can assist clinicians.

2. METHODOLOGY

2.1 Related Work

A substantial body of research has been conducted on CVD prediction using the UCI Cleveland Heart Disease Dataset (CHDD). The following summarises key prior studies that form the baseline for comparison with our work:

Ramakrishnan (2024) applied Random Forest, Decision Tree, Multilayer Perceptron, and XGBoost classifiers on the CHDD and achieved the highest accuracy of 87.28%. Reddy et al. (2024) utilised ten machine learning classifiers spanning categories such as Lazy, Meta, Bayes, Functions, Rules, and Trees, achieving the highest accuracy of 86.4%.

Alotaibi et al. (2025) employed dimensionality reduction, Chi-squared feature selection, and Naive Bayes, achieving a commendable accuracy of 85.15%. Younas (2024) applied six classifier techniques — DT, SVM, KNN, NB, LR, and RF — and achieved the highest accuracy of 86.89% using Logistic Regression. Rani et al. (2025) applied Genetic Algorithm and Recursive Feature Elimination, achieving an accuracy of 86.6%.

Fahim et al. (2025) used a hybrid feature selection algorithm combining Genetic Algorithm with Recursive Feature Elimination alongside KNN, RF, DTs, and XGBoost, achieving 73.72%. Bataineh et al. (2023) implemented a hybrid MLP-PSO algorithm alongside ten diverse ML algorithms, achieving 84.61%. Reddy et al. (2026) evaluated classifiers with and without Principal Component Analysis and achieved 85.8%.

Our study achieves an accuracy of 88.50% using Naive Bayes with SVD feature selection — surpassing all eight prior studies on the same dataset.

2.2 Dataset Description

The experiment was performed on the UCI Cleveland Clinic Heart Disease dataset, a widely acknowledged benchmark for medical machine learning algorithms. The dataset contains data of 303 patients, comprising 207 male and 96 female patients. There are a total of 76 attributes, out of which 13 are selected as prediction features and one is the result attribute.

The result attribute categorises a patient as having CVD (Target = 1) or not having CVD (Target = 0). Of the 303 patients, 176 (Target 1) have CVD and 127 (Target 0) do not have CVD.

Table 1 shows the total number of attributes used in the study:

SR. NO.	Feature	Description
1	Age	Age of patient in years
2	Sex	1 = Male, 0 = Female
3	Chest Pain	Type of chest pain experienced (ranging from mild to severe)
4	Resting Blood Pressure	Normal < 120 mmHg systolic, < 80 mmHg diastolic
5	Cholesterol	High levels cause major cardiac risk
6	Fasting Blood Sugar	Ideal range: 90-130 mg/dL
7	Resting ECG Results	Normal resting heart rate: 60-100 bpm
8	Maximum Heart Rate	Highest beats per minute achieved during exercise
9	Exercise-Induced Angina	Chest pain from insufficient oxygen supply to heart
10	Oldpeak	Measure of ECG abnormality (in mm)
11	Peak Exercise ST Segment	Part of ECG evaluated during exercise stress test
12	Ca (Major Vessels Count)	Ca = 0 indicates presence of heart disease
13	Thal	Nuclear stress test measuring blood flow to heart
14	Target	1 = Has CVD, 0 = Does not have CVD

Table 1: Total Number of Attributes

2.3 Research Methodology

The research methodology followed a systematic pipeline as outlined below:

- Step 1: Load the UCI Cleveland Heart Disease dataset (303 patients, 13 features + 1 target).
- Step 2: Data preprocessing — handle missing values and apply Z-Score Normalisation.
- Step 3: Apply feature selection techniques (SVD and LDA) to reduce dimensionality and noise.
- Step 4: Split dataset into training and test sets using stratified k-fold cross-validation.

- Step 5: Train four classifiers — KNN, Naive Bayes, Logistic Regression, and Decision Tree.
- Step 6: Evaluate classifier accuracy under three conditions: without feature selection, with SVD, and with LDA.
- Step 7: Compare results across all classifiers and feature selection methods and against prior literature.

Accuracy was computed as:

$$Accuracy = (Total\ Number\ of\ Correct\ Predictions / Total\ Number\ of\ Predictions) \times 100$$

2.4 Feature Selection Techniques

Two feature selection techniques were applied to the dataset prior to model training:

Singular Value Decomposition (SVD): SVD is a fundamental matrix factorisation technique that provides a numerically balanced matrix decomposition used for many objectives. It also serves as the underlying algorithm of Principal Component Analysis (PCA). SVD decomposes the data matrix into singular vectors and singular values, capturing the maximum variance in the data in a reduced dimensional space. This removes redundant and correlated features, reducing noise and improving classifier signal quality.

Linear Discriminant Analysis (LDA): LDA is a widely used preprocessing step for machine learning dimensionality reduction. It increases the ratio of between-class inequality to within-class variance, thus confirming maximum class separation. For binary classification tasks such as CVD prediction (yes/no), LDA is particularly powerful as it directly optimises the features for discriminative class boundaries.

3. TOOLS AND TECHNOLOGIES

3.1 Programming Language and Environment

Python 3.x was used as the primary programming language for this project, owing to its extensive ecosystem of machine learning and data science libraries. All experiments were conducted in a Jupyter Notebook environment, which provides an interactive interface for iterative model development, visualisation, and documentation.

3.2 Machine Learning Libraries

The following Python libraries were used for model development and evaluation:

Scikit-learn (sklearn): The primary machine learning library used for implementing all four classifiers — KNN, Naive Bayes, Logistic Regression, and Decision Tree. Scikit-learn also provided utilities for train-test splitting, cross-validation, accuracy scoring, and confusion matrix generation.

NumPy: Used for numerical computation, matrix operations, and implementation of the SVD feature selection technique via `numpy.linalg.svd`.

Pandas: Used for dataset loading, data preprocessing, handling missing values, and feature engineering.

Matplotlib and Seaborn: Used for generating all visualisations including bar charts for accuracy comparison, confusion matrices, and feature correlation heatmaps.

3.3 Feature Selection Tools

Singular Value Decomposition (SVD) was implemented using NumPy's linear algebra module. Linear Discriminant Analysis (LDA) was implemented using the `LinearDiscriminantAnalysis` class from `sklearn.discriminant_analysis`. Both techniques were applied to transform the original 13-feature space into a reduced, discriminative representation prior to model training.

3.4 Dataset Source

The UCI Cleveland Clinic Heart Disease Dataset was obtained from the UCI Machine Learning Repository (DOI: 10.24432/C52P4X). It was originally provided by Janosi, Steinbrunn, Pfisterer,

and Detrano from the Cleveland Clinic Foundation. The dataset is publicly available and has been the standard benchmark for cardiovascular disease prediction research for over two decades.

3.5 Evaluation Metrics

The primary evaluation metric used in this study is Classification Accuracy, defined as the ratio of correct predictions to total predictions, expressed as a percentage. Additionally, Confusion Matrices were generated for qualitative analysis of True Positives, True Negatives, False Positives, and False Negatives. The False Negative (FN) rate was given particular attention given its clinical significance — a false negative represents a missed malignant diagnosis, which can have life-threatening consequences.

4. IMPLEMENTATION

4.1 Data Preprocessing

The UCI Cleveland Heart Disease dataset was loaded using the Pandas library. A preliminary inspection revealed that the dataset contained no significant missing values requiring imputation. The features exhibit different numerical scales — for example, the "Resting Blood Pressure" feature ranges from approximately 94 to 200, while "Fasting Blood Sugar" is a binary (0 or 1) feature. Such disparities in scale can adversely affect distance-based classifiers like KNN.

To address this, Z-Score Normalisation (Standardisation) was applied to all continuous features, transforming each feature to have a mean of zero ($\mu = 0$) and a standard deviation of one ($\sigma = 1$). The normalisation formula is:

$$z = (x - \mu) / \sigma$$

Where x is the original feature value, μ is the mean of the feature, and σ is the standard deviation. After normalisation, the dataset was split into 80% training and 20% test sets using stratified sampling to preserve the class distribution.

4.2 Classifier Descriptions

Four machine learning classifiers were implemented and evaluated:

K-Nearest Neighbor (KNN): The KNN classifier achieves classification by finding the K closest training examples in the feature space and using their labels to determine the class of a query point. It uses the Euclidean Distance formula to compute similarity. KNN is particularly sensitive to the number of features — reducing the dimensionality via SVD or LDA directly reduces the computational complexity from $O(n*d)$ to $O(n*d')$ where $d' < d$, resulting in faster and more accurate predictions.

Naive Bayes (NB): The Naive Bayes classifier uses Bayes' theorem with the strong assumption that features are conditionally independent given the class label. Despite this simplification, Naive Bayes consistently delivers competitive classification accuracy. It computes the posterior probability $P(y|x)$ for each class y given an observation x , and assigns the class with the highest

posterior probability. Its computational simplicity makes it particularly suitable for medical prediction tasks where interpretability and speed are valued.

Logistic Regression (LR): Introduced in the late 1960s as an alternative to Ordinary Least Squares (OLS), Logistic Regression models the relationship between a categorical outcome variable and one or more continuous predictor variables. It is widely used in biomedical research for hypothesis testing and binary classification, making it a natural fit for CVD prediction. The model outputs a probability between 0 and 1, with a threshold of 0.5 used for binary classification.

Decision Tree (DT): Decision Tree is among the most interpretable classification methods. It builds a tree-like structure of decision rules, where each internal node tests a feature, each branch represents an outcome, and each leaf node represents a class label. The tree is constructed by recursively selecting the feature and threshold that maximises Information Gain or minimises Gini Impurity. Decision trees are particularly valuable in clinical settings because they produce human-readable decision paths that clinicians can validate.

4.3 Feature Selection Implementation

SVD Implementation: The dataset matrix X (303 x 13) was decomposed using Numpy's `numpy.linalg.svd` function into matrices U , Σ , and V -transpose. The top k principal components were retained based on the proportion of explained variance, reducing the 13-dimensional feature space to a compact representation that captures the maximum data variance.

LDA Implementation: The `sklearn.discriminant_analysis.LinearDiscriminantAnalysis` class was applied to transform the feature space. LDA projects the data onto a lower-dimensional space that maximises between-class separation and minimises within-class variance. For binary classification (CVD: yes/no), LDA reduces the feature space to a single discriminant component that best separates the two classes.

4.4 Experimental Setup

All four classifiers were evaluated under three experimental conditions for each:

- Without feature selection (using all 13 original features)
- With SVD feature selection (reduced feature representation)

- With LDA feature selection (maximally discriminative projection)

A total of 12 experiments were conducted (4 classifiers x 3 conditions). Each experiment used the same 80/20 train-test split. Results were recorded as percentage accuracy on the test set.

5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

5.1 Classifier Performance Results

Table 2 presents the classification accuracy of all four classifiers under three experimental conditions: without feature selection, with SVD, and with LDA. The results clearly demonstrate that feature selection consistently improves classifier performance across all models.

Classifier	Without Feature Selection	With SVD	With LDA
KNN	62%	85%	86%
Naive Bayes	81%	88.5%	82%
Logistic Regression	81%	85%	83%
Decision Tree	65%	75%	77%

Table 2: Classifier Performance with and without Feature Selection

5.2 Key Observations from Results

Several significant findings emerge from the experimental results:

- Best Overall Performance: Naive Bayes with SVD achieved the highest accuracy of 88.50%, outperforming all other classifier-feature selection combinations.
- Largest Improvement: KNN showed the most dramatic improvement — from 62% (no feature selection) to 86% with LDA, a gain of 24 percentage points. This confirms that KNN is extremely sensitive to dimensionality reduction.
- SVD vs LDA: SVD benefited NB, KNN, and LR most. LDA benefited KNN and DT most. No single feature selection method was universally superior — the optimal choice is classifier-dependent.
- Consistent Improvement: Feature selection improved accuracy for all four classifiers, confirming that the original 13-feature space contained redundant or noisy features that were degrading model performance.
- Decision Tree Benefit: Decision Tree, while the weakest classifier overall (65% baseline), showed consistent improvement with both SVD (75%) and LDA (77%).

5.3 Comparison with Related Work

Table 3 compares our best result (88.50%, Naive Bayes + SVD) against all comparable prior studies on the UCI Cleveland Heart Disease dataset:

Authors	Year	Dataset	Accuracy	vs. Ours
Ramakrishnan	2024	UCI	87.28%	+1.22%
Younas	2024	UCI	86.89%	+1.61%
Rani et al.	2025	UCI	86.6%	+1.9%
Reddy et al.	2024	UCI	86.4%	+2.1%
Reddy et al.	2026	UCI	85.8%	+2.7%
Alotaibi et al.	2025	UCI	85.15%	+3.35%
Bataineh et al.	2023	UCI	84.61%	+3.89%
Fahim et al.	2025	UCI	73.72%	+14.78%
Our Study (NB+SVD)	2026	UCI	88.50%	—

Table 3: Comparison with Related Work Accuracy

The results clearly show that our approach — using a computationally simple Naive Bayes classifier with SVD feature selection — outperforms all prior studies including those employing more complex ensemble methods such as Random Forest, XGBoost, and the hybrid MLP-PSO algorithm. This demonstrates that the right feature selection method matters more than model complexity.

5.4 Feature Selection Analysis

The dramatic performance improvement across all classifiers when feature selection was applied confirms the presence of redundant and noisy features in the original 13-feature dataset. SVD was more effective for probabilistic classifiers (NB, LR) because it transforms features into orthogonal principal components that better satisfy statistical independence assumptions. LDA was more effective for distance-based classifiers (KNN) because it maximises class separation boundaries.

Without feature selection, KNN's 62% baseline accuracy was severely impacted by the curse of dimensionality — all 13 features contributed to the distance computation, including noisy and correlated features that confused the nearest-neighbor calculation. After feature reduction, the distance computation became more meaningful, yielding the dramatic improvement to 86%.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully demonstrated that machine learning can be an effective tool for predicting cardiovascular disease risk from clinical patient data. Four classifiers — KNN, Naive Bayes, Logistic Regression, and Decision Tree — were evaluated on the UCI Cleveland Heart Disease dataset using two feature selection techniques: SVD and LDA.

The key conclusions are as follows:

- Naive Bayes with SVD achieved the highest accuracy of 88.50%, outperforming all eight comparable prior studies on the same dataset.
- Feature selection consistently improved accuracy across all four classifiers, confirming the value of dimensionality reduction for cardiovascular disease prediction.
- KNN showed the largest accuracy gain from feature selection (+24 percentage points with LDA), highlighting the critical impact of the curse of dimensionality on distance-based classifiers.
- The high accuracy and robust performance of Naive Bayes make it a valuable tool for clinicians seeking to predict CVD at an early stage with enhanced success rates.
- A simpler pipeline (NB + SVD) can outperform more complex ensemble methods, demonstrating that appropriate feature engineering is more impactful than model complexity.

6.2 Limitations

While the study achieved strong results, several limitations must be acknowledged:

- Small dataset: The UCI Cleveland dataset contains only 303 patient records, which may limit the generalisability of the results to larger, more diverse patient populations.
- Binary classification: The study addresses only binary CVD prediction (yes/no). It does not distinguish between CVD subtypes, severity stages, or risk gradations.
- Simulation environment: All experiments were conducted offline using a fixed dataset. No real-time clinical validation was performed.

- Limited feature set: The dataset contains only morphological and clinical parameters. Genetic, lifestyle, and socioeconomic factors — all known CVD risk contributors — were not included.

6.3 Future Scope

Several directions for future work have been identified:

- Ensemble Methods: Exploring ensemble classifiers such as Random Forest, Gradient Boosting, and XGBoost stacking with optimised feature selection could potentially push accuracy beyond 90%.
- Larger Datasets: Incorporating larger, multi-centre datasets would improve the generalisability and clinical reliability of the model.
- Deep Learning: Investigating deep learning architectures such as LSTM (for sequential patient data) and feedforward neural networks with attention mechanisms could capture non-linear risk patterns.
- Explainable AI (XAI): Integrating SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) to produce clinical heatmaps and feature contribution reports would enhance physician trust in model decisions.
- Multi-class Extension: Extending the framework to multi-class classification — distinguishing not only whether CVD is present, but its type and severity stage — for more actionable clinical guidance.
- Real-Time Deployment: Developing a web-based or mobile application for real-time CVD risk prediction that clinicians can use at the point of care.

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